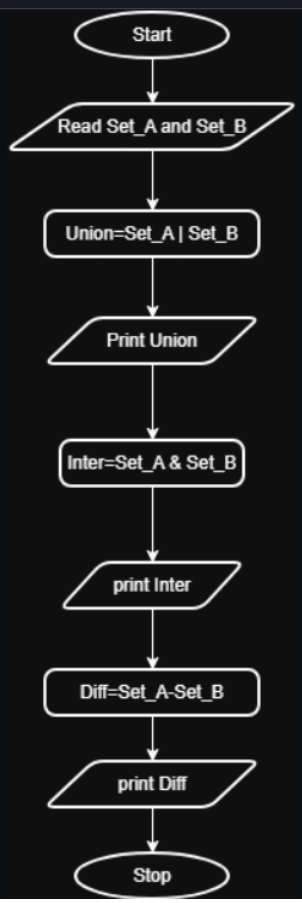
Algorithm

1. Start.
2. Declare two sets A and B
3. Read elements of Set A from the user
4. Read elements of Set B from the user
5. Perform Union operation Union = A ∪ B.
6. Perform Intersection operation Intersection = A ∩ B.
7. Perform Difference operations
8. Difference1 = A – B.
9. Display Set A and Set B
10. Display Union, Intersection, and Difference results
11. Stop



CODETANTRA Home

ayush.dehankar.batch2025@sitnagpur.siu.edu.in Support Logout

Programming and Problem Solving Lab - TE7287 - II Sem - 2026

Pin

Search course ctrl + k

1. Experiment - 1
2. Experiment - 2
3. Experiment - 3
4. Experiment - 4
4.1. Programs
4.1.1. Set Operations
5. Experiment - 5
6. Experiment - 6
7. Experiment - 7
8. Experiment - 8
9. Experiment - 9
10. Experiment - 10
11. Experiment - 11
12. Experiment - 12

4.1.1. Set Operations

08:28

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat...

Submit

Debugger

```python
set_a=set(map(int,input("Set A: ").split()))
set_b=set(map(int,input("Set B: ").split()))
union = set_a | set_b
print("Union:", union)
Inter = set_a & set_b
print("Intersection:", Inter)
Diff = set_a - set_b
print("Difference:", Diff)
```

Terminal Test cases

< Prev Reset Submit Next >